

MATTHEW DAVIES

davies62@uwindsor.ca | (519)-312-0635 | linkedin.com/in/matthew-davies | github.com/davo300

AI/ML Intern | Agentic AI, Data Pipelines, Python Systems

EDUCATION

University of Windsor

Windsor, ON

Honours Computer Science — AI Specialization — Mathematics Minor

Sep. 2022 – Aug. 2026

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, Golang, C++, C, C#, PHP

Web / Full-Stack: FastAPI, Spring Boot, Node.js, React, REST APIs, CSS, MySQL, PostgreSQL, Laravel

Automation / Tools: Selenium (Python), Git, GitHub, Linux, Bash/Shell Scripting, Postman, JUnit, xUnit, JavaFX

Cloud / DevOps: AWS (EC2, RDS, Elastic Beanstalk), Docker, Containers, Procfile Deployment, CI/CD

Data & Analytics: Retrieval-Augmented Generation (RAG), SQL, FAISS, embeddings, ETL pipelines, Pandas, NumPy, Matplotlib, Seaborn, Tableau, Excel

Systems / Performance: CUDA, GPU Programming, Parallel Computing

Relevant Courses: Big Data Analytics (Apache Spark), Machine Learning, Probability & Statistics, Database Management Systems (SQL), Numerical Analysis (Matlab), Design & Analysis of Algorithms (Proofs).

PROJECTS

CS Tutor Bot — RAG-Based Course Assistant

- Built a **Retrieval-Augmented Generation (RAG)** data platform supporting agentic workflows for ingesting, indexing, and querying large document datasets.
- Implemented data ingestion, embedding generation, and semantic retrieval using **MiniLM + FAISS**.
- Deployed a FastAPI backend to support **125+ students** with source-aware, non-hallucinating responses.
- Built and validated end-to-end **data pipelines** for ingestion, indexing, retrieval, and response generation, emphasizing correctness, reliability, and containerized deployment.

GPU-Accelerated Embedded Systems — CUDA & NVIDIA Jetson

- Built **GPU-accelerated systems** spanning **embedded robotics** and **numerical simulation** for real-time and high-throughput workloads.
- Developed an **autonomous JetBot** on **NVIDIA Jetson Nano**, integrating real-time vision pipelines in **Python/OpenCV** under compute constraints.
- Implemented a **2D heat diffusion solver** in **C/CUDA** using the Jacobi method, optimizing kernel launches, memory access, and synchronization.
- Benchmarked **GPU vs CPU** execution on NVIDIA hardware, analyzing memory bandwidth, occupancy, and performance tradeoffs.

Machine Learning Strategy Classifier — Iterated Prisoner's Dilemma

- Built a supervised ML pipeline to classify Iterated Prisoner's Dilemma strategies using behavioral features.
- Trained and evaluated Decision Tree, Random Forest, and Gradient Boosting models, achieving high classification accuracy while analyzing feature importance and generalization behavior.
- Performed statistical analysis and visualization using **Pandas, Matplotlib, and Seaborn**.
- Generated labeled datasets from simulation runs and evaluated model performance using cross-validation.

Google Docs API — Data Extraction & ETL Pipeline

- Developed a **Python ETL pipeline** using the **Google Docs API** to extract, transform, and load structured data from a private document.
- Parsed and validated semi-structured table data, handling inconsistent formats and ensuring data integrity.
- Transformed coordinate-based symbol data into a 2D grid to reconstruct and visualize a decoded message.

EXPERIENCE

NASA Space Apps Challenge

Windsor, ON

Software Developer — Hackathon

Oct. 2025

- Worked in a multidisciplinary team to build an interactive web platform for NASA's high-resolution imagery.
- Implemented UI functionality and image manipulation tools using **React (TypeScript)** and **Go**.
- Collaborated with cross-functional teammates to translate user needs into data-driven UI and backend functionality.

Google Developer Society

Windsor, ON

Team Member

Apr. 2024 – Present

- Mentored students on programming practices, debugging, and software tooling.
- Collaborated on workshops and contributed to a community of student developers.